EXPLAINABILITY IN HYPERSPECTRAL IMAGE CLASSIFICATION: A STUDY OF XAI THROUGH THE SHAP ALGORITHM

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ABSTRACT

Hyperspectral Imaging (HSI) captures complex spectral signatures across a broad range of electromagnetic wavelengths, offering diverse applications from remote sensing to medical diagnostics. HSI classification assigns labels to pixels in HSI cubes, traditionally using handcrafted features and, more recently, by deep neural networks (DNNs). However, DNNs' inherent lack of interpretability has driven the growing adoption of eXplainable AI (XAI) techniques, aimed at explaining the decision-making process. This study integrates the SHapley Additive exPlanations (SHAP) algorithm with a convolutional neural network (CNN) for HSI classification, facilitating a per-patch understanding of class contribution. Validated using the Indian Pines dataset, the framework enhances understanding of class roles in HSI classification through XAI.

***Index Terms*—** Hyperspectral Imaging (HSI), Classification, Convolutional Neural Network (CNN), eXplainable Artificial Intelligence (XAI), SHAP (SHapley Additive exPlanations)

1. INTRODUCTION

Hyperspectral Imaging (HSI), also known as spectroscopy, utilizes a broad range of electromagnetic wavelengths (consisting of several hundreds of narrow and contiguous channels) to capture detailed spectral signatures from objects or scenes. HSI sensors, typically operating within the 0.4 to 2.5 µm spectral region, excel in addressing challenges beyond the capabilities of multispectral or natural images, allowing precise pixel-level identification [1]. Hyperspectral remote sensing generates hyperspectral images that reflect the unique physical properties of ground features [2] and finds diverse applications, including but not limited to smart agriculture [3], oil spill detection [4], and urban planning [5]. Moreover, HSI proves useful in medical contexts [6] and food quality evaluation [7] while its sensors are applicable in space, air, and underwater vehicles [8].

HSI classification is one of the main research areas in the hyperspectral community and has received considerable attention during the past several decades. The primary objective in HSI classification is to effectively assign specific labels to individual pixels within the HSI cube, leveraging the combined information of spectral and spatial properties. Conventional machine learning techniques for HSI classification tasks were reliant on the extraction of handcrafted features. While these methods exhibit effectiveness up to a certain degree, they might not entirely capture the complex nature of hyperspectral data. Consequently, recent progress in the field of computer vision has steered the focus toward deep neural networks (DNNs) for the automatic extraction of features directly from the raw data thereby improving classification performance. Notably, convolutional neural networks (CNNs), have shown remarkable success in various HSI classification tasks as well as multimodal remote sensing data [9]. Furthermore, state-of-the-art studies have addressed topics such as spectral variability, land use and land cover classification, and anomaly detection through the application of deep learning or artificial intelligence (AI) models in HSI [10, 11, 12].

One of the primary challenges associated with DNNs is their intrinsic lack of interpretability. While DNNs have shown impressive performance in a variety of tasks, their complex architectures and multitude of parameters make it challenging to understand the underlying mechanisms that drive their decision-making process. This lack of transparency frequently leads to DNNs being characterized as "black boxes," and it raises concerns, especially in applications where understanding the rationale behind a model's outcome is important for building trust and responsibility. To address this problem, researchers have been actively investigating various techniques in eXplainable Artificial Intelligence (XAI) to improve the interpretability and explainability of DNNs. It's worth noting that the terms "explainability" and "interpretability" are frequently used interchangeably in the literature, while some studies introduce subtle distinctions between these notions. One seminal method in the XAI domain, which has received a lot of attention, is called SHAP (SHapley Additive exPlanations) [13]. Exploiting a game theoretic approach, SHAP, gives insights into the contribution of each feature to a model's output. Recently, few investigations on XAI applied to HSI have been undertaken [14, 15, 16].

In this paper, we utilize the SHAP algorithm alongside a CNN to provide explanations for HSI classification. Adopting the methodology introduced by [17], we applied the principal component analysis (PCA) to perform dimensionality reduction on the spectral dimension of the HSI data cube. Subsequently, we examine the contribution of each class to the classification results for individual patches. Put simply, the SHAP algorithm is applied on a per-patch basis to tensors of dimensions $s \times s \times cr$, where cr represents the number of principal components and s indicates the number of neighboring pixels. We validated our proposed framework using the Indian Pines image, which was acquired using the AVIRIS sensor to cover agricultural areas in Northwestern Indiana [18]. The main contribution of this work is to enhance our understanding of class contributions in HSI classification through the use of XAI techniques. This is essential in the field of remote sensing, where not only accurate classification is sought, but also understanding "why and how" the results are achieved is crucial for building trust in decision-making processes. Furthermore, this research serves as a stepping stone for future investigations into XAI, aiming to enhance not only interpretability but also classification accuracy in hyperspectral analysis, especially when dealing with limited training data.

The remaining parts of this study are organized as follows: Section 2 provides a concise introduction to the SHAP algorithm. In Section 3, the proposed framework is described. In Section 4, the real data analysis is presented. Lastly, Section 5 presents the conclusions drawn from this study.

2. SHAP EXPLANATIONS FOR IMAGE CLASSIFICATION

SHAP (SHapley Additive exPlanations) [13] is a method for explaining the output of complex machine learning models, in our case CNNs. It attributes the contributions of input features to the model's predictions to facilitate the understanding of how different parts of the input image impact the model's decision. The key concept behind SHAP is to assign the Shapley value to each input feature (in this scenario, pixels in an image) based on the contribution to the model's prediction. Shapley values are inspired by coalitional game theory to ensure a fair distribution among the features [19]. The Shapley value for a specific pixel i in an input image can be given by:

$$\varphi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N|-|S|-1)!}{|N|!} \{f_{S\cup\{i\}}(x_{S\cup\{i\}}) - f_S(x_S)\} \tag{1}$$

where N is the set of all features (pixels) in the image, and $S \subseteq N \setminus \{i\}$ is a subset of features excluding i. Additionally, it involves training two surrogate models: one with the i-th pixel included ($f_{S\cup\{i\}}$) and the other with the i-th pixel excluded (f_S). Subsequently, predictions from both models are compared for the current input, calculated as $f_{S\cup\{i\}}(x_{S\cup\{i\}}) - f_S(x_S)$, where x_S represents the values of the input features within the set S. Subsequently, the SHAP explanation can be formulated as:

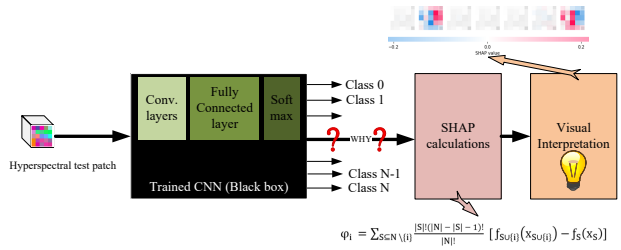


Fig. 1: High-level architecture of the proposed framework

$$g(z') = \varphi_0 + \sum_{i=0}^{M} \varphi_i z'_i \tag{2}$$

where g represents the explaining model, $z' = \{0,1\}^M$ denotes the simplified feature representation with 1 indicating the presence and 0 indicating the absence of a feature. M shows the number of simplified input features. By computing SHAP values for all pixels in an image, one can visualize which parts of the image positively or negatively affect the classification outcome. This offers insights into CNN's decision-making mechanism for each individual test image.

3. THE PROPOSED FRAMEWORK

In this section, we present the proposed framework that integrates the SHAP algorithm with a CNN to enhance the interpretability of HSI classification. The objective is to illuminate on how each pixel contributes to the predictions made by the CNN, particularly on a patch-wise level. The high-level architecture of the proposed framework is shown in Fig. 1. The framework consists of the following key components: CNN model, SHAP analysis, and interpretation visualization. The CNN model is responsible for classifying HSI patches into different classes. The model is trained using a dataset containing patches of hyperspectral data along with their corresponding labels. Section 4.1 provides more details on the CNN and its implementation. After training the CNN model, SHAP analysis is performed on individual patches. In the end, computed SHAP values are transformed into visual representations to provide an understanding of class contributions and the influence of various spectral bands on HSI classification.

4. REAL DATA ANALYSIS

4.1. CNN

In this paper, we employ a very simple CNN that consists of only two convolutional filters and one fully connected layer.

Table 1: CNN architecture

Layer	Name	Input size	Output size	Activation Function	Number of Trainable Parameters
0	Input	None×5×5×30	None×5×5×30	-	-
1	Conv 1	None×5×5×30	None×3×3×64	ReLU	17344
2	Conv 2	None×3×3×64	None×1×1×32	ReLU	18464
3	Flatten	None×1×1×32	None×32	-	-
4	FC	None×32	None×16	Softmax	528

The architecture of the CNN is depicted in Table1. The convolutional filters have a kernel size of 3×3, with no padding added to the input before performing the convolution operation, and hence a reduced output size. The total number of trainable parameters is 36,336. The model is compiled using the Adam optimizer and the categorical cross-entropy loss function, implemented by the Keras library backend for TensorFlow version 2.10.0.

4.2. DATASET FORMATION

The Indian Pines image was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the Indian Pines test site in North-Western Indiana [18]. It includes 224 spectral bands within a wavelength range of 400–2500 nm, though the number of bands is reduced to 200 by removing bands covering the region of water absorption. The spatial dimensions of the image are 145 × 145 pixels, with a spatial resolution of 20 m per pixel (MPP). The dataset includes sixteen distinct vegetation classes, covering agricultural fields with various crops, as well as sections of forests and dense vegetation. This dataset is a significant reference for the research community, widely used for evaluating classification algorithms, though some rare classes are often excluded from evaluations due to their limited representation. Indian Pines image is shown in Fig. 2. Similar to [17], we apply PCA to reduce the spectral dimensions of the HSI data cube from (145, 145, 200) to (145, 145, 30). Employing a window size of 5, a patch dataset with dimensions of (10249, 5, 5, 30) is generated and divided into train and test datasets. In the end, through the oversampling of underrepresented classes and integrating data augmentation techniques to the training images, a training dataset of size (29865, 5, 5, 30) is formed, while the testing dataset has dimensions of (2563, 5, 5, 30). Further information regarding the implementation can be found in [20].

4.3. RESULTS

Following the training phase, the model is evaluated on the test dataset, and the non-normalized confusion matrix and classification performance metrics are shown in Fig 3. The term "support" in the last column of Fig. 3 (b) refers to the number of test patches. Subsequently, patches with the same size as the training dataset (5×5×30) are extracted from the

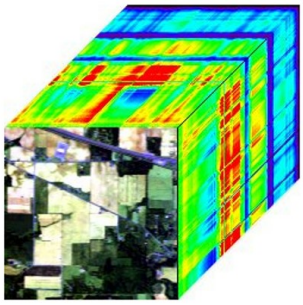
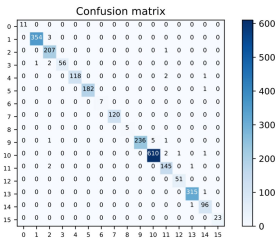


Fig. 2: Indian Pines hyperspectral data cube



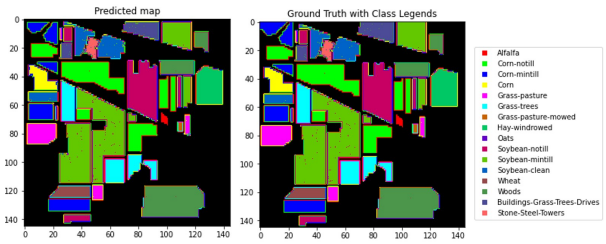
	precision	recall	f1-score	support
Alfalfa	1.00	1.00	1.00	11
Corn-notill	1.00	0.99	0.99	357
Corn-mintill	0.96	1.00	0.98	208
Corn	1.00	0.95	0.97	59
Grass-pasture	1.00	0.98	0.99	121
Grass-trees	1.00	0.99	1.00	183
Grass-pasture-mowed	1.00	1.00	1.00	7
Hay-windrowed	1.00	1.00	1.00	120
Oats	1.00	1.00	1.00	5
Soybean-notill	1.00	0.97	0.99	243
Soybean-mintill	0.99	0.99	0.99	614
Soybean-clean	0.96	0.98	0.97	148
Wheat	0.98	1.00	0.99	51
Woods	0.99	1.00	1.00	316
Buildings-Grass-Trees-Drives	0.96	0.99	0.97	97
Stone-Steel-Towers	1.00	1.00	1.00	23
accuracy			0.99	2563
macro avg	0.99	0.99	0.99	2563
weighted avg	0.99	0.99	0.99	2563

(a) (b)

Fig. 3: Evaluation of the test dataset (a) confusion matrix and (b) classification performance

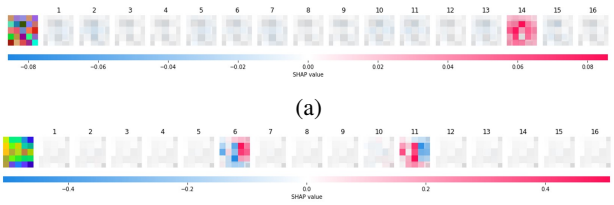
image (145×145×30) and fed to the CNN for prediction. Although the analysis of borders is not performed, the model provides an overall accuracy of 99.09%. The predicted image is shown in Fig. 4 (a), along with the ground truth for reference Fig. 4 (b).

Moving forward, we select a few patches for SHAP analysis. For instance, consider the patch centered at coordinates (x,y)=(100,130), where the network accurately predicts "woods" (class 14). This specific patch and its SHAP values are depicted in Fig. 5 (a). Similarly, for the patch centered at coordinates (x,y)=(115,50), where the network correctly identifies "Soybean-mintill" (class 11), the associated patch and its SHAP values are shown in Fig. 5 (b). In Fig. 5 (a), most pixels associated with class 14 are highlighted in red, representing positive SHAP values. These positive values signify pixels that have a positive impact on the classification outcome. In contrast, in Fig. 5 (b), while the correct class (class 11) predominantly shows positive SHAP values,



(a) (b)

Fig. 4: Experiments with Indian Pines dataset, (a) the predicted image and (b) the ground truth

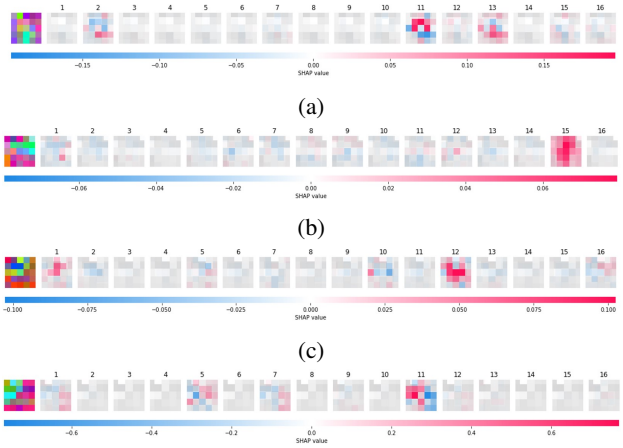


(a)

(b)

Fig. 5: SHAP values across 16 different classes for the patch centered at coordinates (a) (x,y)=(100,130) and (b) (x,y)=(115,50) The actual and predicted classes are (a) class 14 "woods" and (b) class 11 "Soybean-mintill".

the alternate class 6 (Grass-trees) is characterized by dominantly negative values depicted in blue, which supports the accuracy of the prediction. Let's examine some examples where the network's decisions are incorrect. For instance, the patches centered at coordinates (x,y)=(25,94), (x,y)=(27,94), (x,y)=(29,94), and (x,y)=(30,94) are misclassified as classes, 13, 15, 12, and 5, respectively, whereas their true label is class 11. The SHAP values for these cases are depicted in Fig. 6 (a) to (d). Note that each patch possesses dimensions of (5×5×30), and SHAP yields a list of (16×5×5×30) values, with 16 denoting the number of classes. However, in Fig. 5 and Fig. 6, the patches and per-class SHAP outputs are visually represented in 2D rather than 3D. To enhance clarity, let's examine the example in Fig. 6 (a). By computing the spatial mean (averaging over 5×5 pixels) of absolute SHAP values, we can illustrate the average SHAP value per class and per spectral band (16×30) as depicted in Fig. 7. As previously mentioned in relation to Fig. 6 (a) and Fig. 7, the true class is 11, whereas the network's prediction is class 13. Regardless of specific classes, it is evident that certain spectral bands have a more significant impact on the decision-making process. Note that in Fig. 7, due to the use of the absolute operation, positive and negative contributions are combined. To elaborate, the variation of SHAP values across spectral bands for the example illustrated in Fig.6 (a) is shown in Fig. 8 where in alignment with Fig. 7, a significant positive contribution in the spectral band #4 can be observed.



(a)

(b)

(c)

(d)

Fig. 6: SHAP values across 16 different classes for the patches centered at coordinates (a) (x,y)=(25,94), (b) (x,y)=(27,94), (c) (x,y)=(29,94), and (d) (x,y)=(30,94). These patches are predicted as classes 13, 15, 12, and 5, respectively, while all of them belong to class 11 "Soybean-mintill"

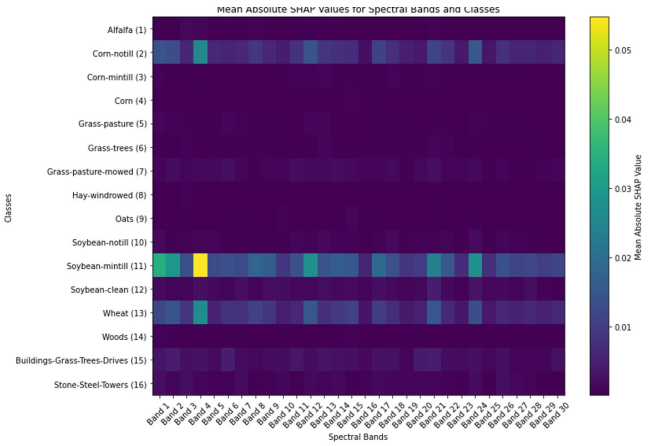


Fig. 7: Average absolute SHAP values across classes and spectral bands for the example illustrated in Fig. 6 (a)

5. CONCLUSION

In this study, we utilized the SHAP algorithm to enhance the interpretability of CNN-based HSI classification. Through a per-patch SHAP analysis, we investigated correct and incorrect predictions by illustrating feature importance and highlighting the impact of spectral bands. This study establishes a pathway for future research, where XAI techniques can improve not only interpretability but also accuracy in HSI classification, particularly with limited training data. As a potential future research direction, ablation test, computational complexity, and noise effects can be analyzed. Moreover, XAI can be applied to multimodal remote sensing data for applications such as oil spill detection [21, 22].

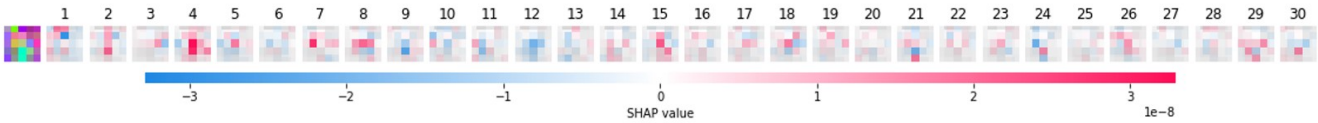


Fig. 8: Variation of SHAP values across spectral bands for the example illustrated in Fig. 6 (a)

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